

Summary of "Power Stabilization for AI Training Datacenters"

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Problem and Motivation

Large-scale AI training now spans tens or even hundreds of thousands of GPUs, producing highly synchronized cycles of compute-intensive and communication-intensive phases. These phases cause rapid and repeated swings between near-TDP power draw and near-idle power. When thousands of GPUs act in synchrony, these fluctuations amplify into datacenter-scale oscillations that can reach tens or even hundreds of megawatts. More critically, the oscillation frequencies often fall within the sub-synchronous range (0.1–3 Hz), which overlaps with natural resonances of turbine-generator shafts and long transmission lines. If these frequencies align, they can damage grid infrastructure or trigger instability events. As AI workloads continue growing, uncontrolled power swings are becoming a key bottleneck for safe and scalable deployment.

Related Works

Prior work has made efforts on parallelism strategies for large-scale model training such as Megatron-LM, the growth of foundation models like GPT-3, PaLM, and Llama, and asynchronous training techniques that reduce synchronization needs. Grid-side research has explored sub-synchronous resonance, torsional instability in turbine generators, and inter-area oscillations across large transmission networks. Hardware documentation (e.g., NVIDIA GB200 guides) outlines GPU power behavior, including EDP peaks and ramping characteristics. Combined, these works highlight the intersection between massive AI workloads and power system vulnerabilities, but none provide an integrated, cross-stack solution tailored to hyperscale synchronized jobs.

Solution Overview

To address the severe power swings generated by hyperscale AI training workloads, the paper proposes a **three-layer mitigation strategy** that spans **software, GPU hardware, and datacenter infrastructure**. Each layer targets different aspects of the power waveform, and together they provide a coordinated response capable of meeting diverse utility specifications.

Software-Only Mitigation (Firefly)

The first mitigation layer is a software-based smoothing mechanism that dynamically injects secondary workloads to maintain a stable power draw. When the primary job enters communication or synchronization phases, the system launches synthetic compute kernels (e.g., GEMM operations) to sustain GPU activity near a preselected target.

GPU power smoothing (e.g., NVIDIA GB200): Allows programmable ramp rates, minimum power floors, and stop delays. This is more reliable and has lower overhead, but MPF ceilings ($\leq 90\%$ TDP) limit its ability to meet strict utility specifications.

This approach relies on **fast GPU telemetry**, particularly block-activity counters that reflect real-time GPU usage. Since reliable hardware power counters often operate at 100 ms granularity, too slow for meeting frequency-domain requirements, the system instead reads activity counters at around 1–5 ms intervals. Based on these readings, Firefly decides when to activate or deactivate the artificial workload.

Key behavior includes:

- **Start condition:** When power or block activity falls below a threshold, Firefly immediately launches the secondary workload to prevent a sharp drop.
- **Stop condition:** The secondary workload backs off whenever the primary job ramps up again; because GPU counters are not per-process, Firefly must periodically “probe” activity by briefly yielding resources.
- **Control mechanism:** Firefly uses NVIDIA’s Multi-Process Service (MPS) to share GPU contexts, allowing a secondary workload to coexist with the training job.

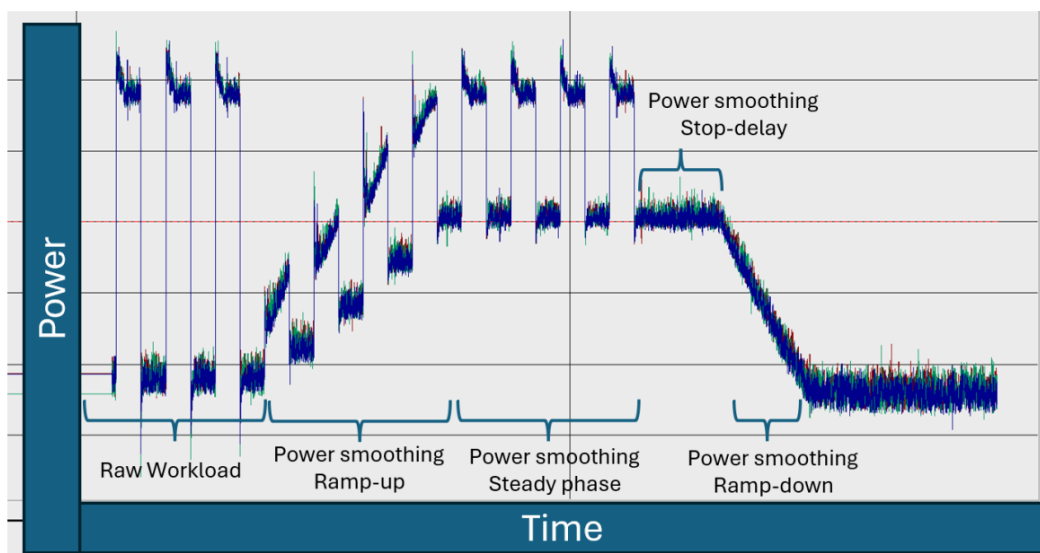


Fig. 5. GB200 Power smoothing results with a square-wave microbenchmark.

While flexible and deployable without hardware changes, this method incurs several drawbacks: performance overhead (<5%), CPU and bandwidth costs from fast telemetry ingestion, reliability concerns due to shared GPU contexts, and wasted energy if the secondary workload does no useful work.

GPU-Level Power Smoothing

To overcome the limitations of software smoothing, the second layer uses **hardware-supported power control**, recently introduced in devices such as the NVIDIA GB200. This mechanism allows cloud providers and developers to program **power profiles** directly into the GPU's firmware:

- **Ramp-Up and Ramp-Down Rates:** These directly enforce utility time-domain constraints by limiting how quickly GPU power may increase or decrease (in W/s).
- **Minimum Power Floor (MPF):** Ensures the GPU never drops below a configurable fraction of its TDP during idle or communication phases. This reduces high-frequency components in the power spectrum.
- **Stop Delay:** Determines how long the GPU remains at MPF after workload activity ceases.

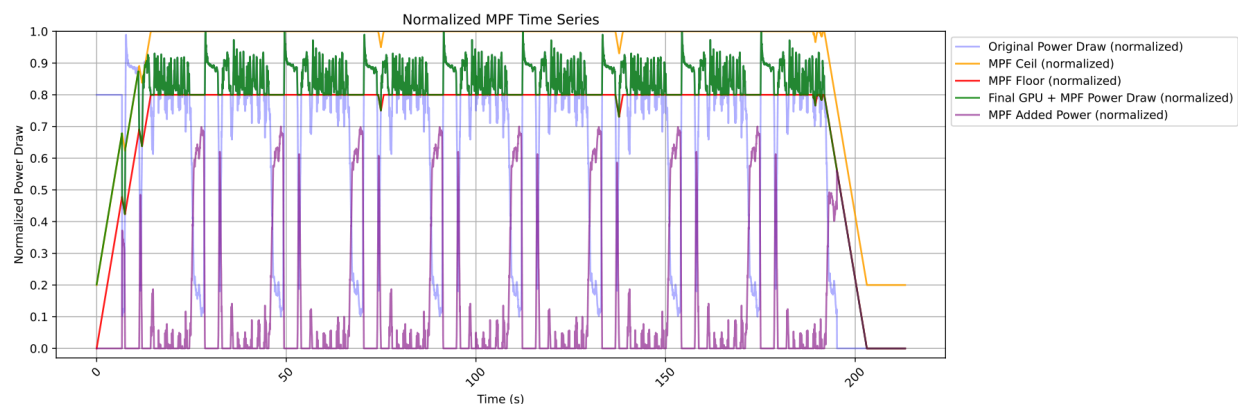


Fig. 6. Power smoothing to the minimum power floor (MPF) simulated on the training waveform from Figure 1.

These settings allow very fast transitions because they operate inside the GPU's own power controller, making them suitable for meeting frequency-domain specifications as well.

A simulation using Microsoft's StratoSim (Fig. 6 above) demonstrates that setting an MPF around 90% TDP significantly reduces oscillation amplitude, though at the cost of ~10.5% increased energy consumption.

Rack-Level Energy Storage

The third layer introduces a **rack-level DC energy-storage system** capable of absorbing and releasing power to offset rapid swings. This solution directly shapes the power waveform without

generating additional energy waste, making it the most effective mitigation for amplitude reduction.

Why Rack-Level:

- Rack-level placement balances proximity and scalability:
 - Close enough to respond within milliseconds
 - Far enough to reduce per-unit cost and avoid touching every server
- AC–DC converters already exist at the rack boundary, simplifying integration
- Avoids exposing upstream devices (PDUs, UPSes) to sharp transients
- Failure of individual rack-level units does not propagate datacenter-wide

Operational Principle:

During communication phases, the storage **charges**, absorbing unused power. During compute phases, it **discharges**, supplementing the rack's instantaneous demand. The simulation in Figure 7 shows how this approach flattens the power waveform while keeping the storage's state of charge within manageable bounds.

Challenges:

- Very low-frequency (<1 Hz) fluctuations require extremely large capacitance
- High costs (capex + embodied carbon) if designed to cover rare worst-case events
- Requires co-design with GPU-level smoothing to manage extreme ramp transitions

Combined Architecture

Solution	Reliability	Performance	Energy	Cost	Ability to meet tightest spec	Dependency on the developer	Lifetime
Software-only mitigation	Medium	Medium	High	Medium	High	High	High
GPU power smoothing	High	Medium	High	Low	Medium	Medium	Medium
Rack-level energy storage	High	High	Low	High	High	Low	High

TABLE I

SUMMARY OF VARIOUS PROPOSED SOLUTIONS. FOR ENERGY, COST, AND DEPENDENCY ON THE DEVELOPER, LOWER IS BETTER.

Because no single approach satisfies all utility specifications across both time and frequency domains, the paper proposes a **hybrid approach**:

- **GPU-level smoothing** handles short-term ramp constraints and fast-frequency components.
- **Rack-level energy storage** absorbs deeper power fluctuations without wasting energy.
- **Software-based smoothing** acts as a fallback when:
 - utility dynamic range limits exceed MPF capabilities
 - energy storage temporarily saturates or drains
 - workloads exhibit unpredictable patterns

Benefits:

- Meets most stringent grid requirements
- Minimizes wasted energy
- Distributes responsibilities across layers
- Reduces reliance on per-job tuning
- Provides reliable long-term operation at hyperscale

The hybrid solution is considered optimal for upcoming training clusters that may reach or exceed 100,000 GPUs and operate near grid infrastructure limits.

Limitations

Although effective, these solutions carry trade-offs. Software smoothing wastes energy, competes for GPU/CPU resources, and can reduce reliability because primary and secondary workloads share the same GPU context. GPU-level smoothing is constrained by hardware limits, especially the maximum MPF, preventing full compliance with stringent dynamic-range specifications. Energy-storage systems require significant cost, space, and embodied carbon, and achieving sufficient capacitance to handle low-frequency swings is challenging. Moreover, all approaches rely on fast and accurate telemetry; without it, detection and mitigation of harmful frequencies are delayed. Thus, none of the strategies individually solves the problem for all utility environments.

Future Research Directions

Future work should explore power-aware and less-synchronous training algorithms that inherently reduce load variability without degrading convergence. Improved coordination between utilities and datacenter operators is needed to standardize resonance and ramp specifications. Cross-industry collaboration—potentially through organizations like OCP—could establish shared standards for telemetry, signaling, and sub-synchronous oscillation mitigation. Advancing long-duration battery energy storage systems (BESS) and integrating them with rack-level systems could support even larger training workloads. Finally, more robust real-time spectral monitoring and automated backstop systems could detect and suppress emerging instabilities before they propagate.

Summary of Class Discussion

Question 1: Firefly offers several software mitigations for GPU workloads and power spikes, but they add performance overhead and extra energy use. What are its benefits compared to the other solutions?

Answer: Firefly's big advantage is that it is pure software, so it can actually be deployed today without waiting for new hardware or power infrastructure. On top of that, it can work together

with future hardware features, enabling better scheduling and GPU level power control, so it is flexible and can evolve instead of being a one-off fix.

Question 2: Are any companies working on rack-level energy storage, and what are the downsides vs big batteries like Tesla Megapack? Does storage just delay the problem, and what if the batteries are “full”?

Answer: Rack-level storage is still mostly a research idea, not something widely deployed. You still need a huge amount of batteries either way, plus extra complexity to cool and manage many small units. In practice, both rack-level and big BESS systems try not to hit “full” by using prediction and control software and keeping some headroom, so the batteries can absorb spikes and then slowly ramp down instead of just postponing a crisis.

Question 3: Do different accelerators or GPU generations change the power-fluctuation problem in a meaningful way?

Answer: Different chips draw power a bit differently, but unless they are massively more efficient, you still need lots of them, so big swings in power do not just disappear. Mixing very different chips in the same rack can even make control harder, which is why most ideal designs assume fairly uniform hardware so scheduling and power control are easier.

Question 4: Why do companies care about asynchronous training and advanced scheduling here?

Answer: Asynchronous training and smarter scheduling mainly stop one slow step or flaky link from stalling the whole job, so they boost performance. As a side effect, they also break up perfectly synchronized compute and communication phases, which tends to spread out power use over time and smooth some of the sharp spikes.

Question 5: Why is it hard for companies to adopt academic power-management systems directly?

Answer: To really use those systems, you usually have to touch many layers at once, it is a big engineering and reliability risk. Vendors also like to rebuild ideas in their own stack, so academic prototypes rarely get dropped in exactly as they are, even if the ideas influence what they eventually ship.

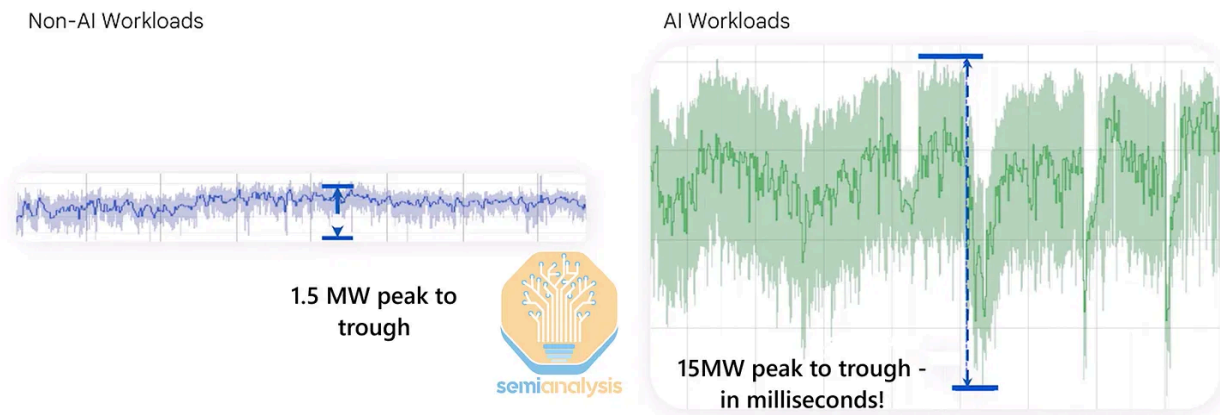
Question 6: How much memory does Bitcoin mining use, and could Bitcoin-style workloads help smooth data-center power usage?

Answer: Bitcoin mining is mostly pure compute and does not use much memory, so it can be turned on and off quickly. In theory, that makes it a nice filler workload to soak up spare power instead of running useless dummy jobs, but in practice money, regulation, and integration issues mean it is more of a thought experiment than a realistic near-term fix.

Summary of "AI Training Load Fluctuations at Gigawatt-scale – Risk of Power Grid Blackout?"

Problem and Motivation

AI training workloads draw massive power (hundreds of megawatts to gigawatts) and, importantly, their load profile is highly dynamic: they can swing from near full load to near idle in fractions of a second.



Existing power grids were designed for relatively smooth, predictable large loads (industrial, data centers, etc.). They are not designed for extremely fast fluctuations in demand at this scale.

These fluctuations introduce risks: grid frequency/voltage stability is challenged when supply-demand balance is upset on very short timescales (milliseconds to seconds).

Specifically, the article points out that in regions such as the Electric Reliability Council of Texas (ERCOT), there is a queue of ~108 GW of large loads (many datacenters) seeking connection, which increases the systemic risk.

Scenario (Load Loss Level - MW)		Status
~3,600		Insecure - Wide Area Voltage Issues
~2,600		Insecure - Wide Area Voltage Issues
~2,450		Insecure - Wide Area Voltage Issues
~2,100		Insecure - Local Voltage Issues
~2,000		Insecure - Local Voltage Issues
~1,950		Secure

The motivation is that if a large AI training datacenter load drops offline (or quickly swings) it could trigger cascading grid failures or even blackouts through frequency/voltage excursions.

Related Works

The article references how grid frequency and voltage stability depend on supply = demand balance, and how conventional generators provide system inertia to buffer rapid fluctuations.

It also references specific fault-and-load-shedding studies in ERCOT: for example, presentations modelling what happens when gigawatts of datacenter load disconnect rapidly and how that affects grid frequency, rate of change of frequency (ROCOF), etc.

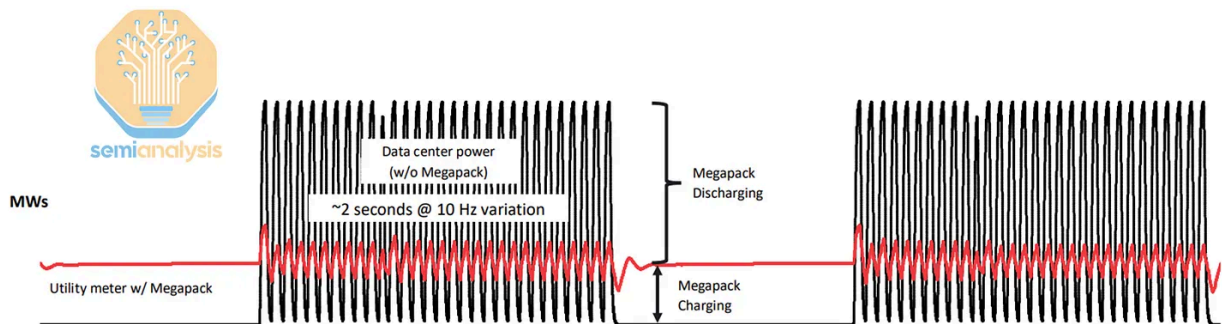
Solution Overview

It emphasises that purely software or scheduling fixes (e.g., smoothing workload start/stop, staggered synchronization) are necessary but unlikely to fully resolve the issue at gigawatt scale. The authors note that at least some “hardware-based” interventions will be required.



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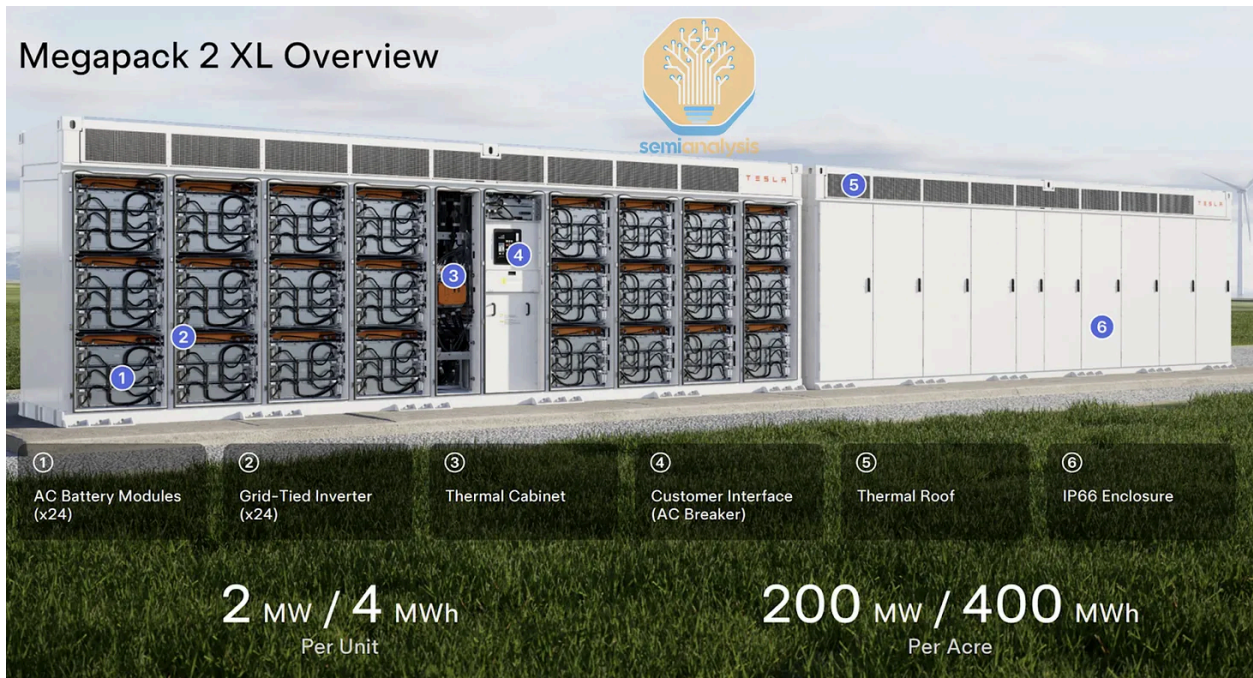
Solution: Load Smoothing with Megapack can reduce 70%+ of variability



Connecting Megapack in parallel to the load helps reduce variability → Improves grid reliability & power quality

- Energy throughput modeling shows 20+ year lifetime
- Charging and discharging are balanced such that BESS SOC is maintained for a 24/7 smoothing operation

Among hardware solutions they highlight: battery energy storage systems (BESS) like Tesla Megapack installations, supercapacitors, synchronous condensers at the interconnect point, active load smoothing devices.



They also discuss cost/scale trade-offs: for example, a synchronous condenser installation for a 1GW datacenter may cost in the tens of millions of dollars.

Limitations

The analysis involves modelling and scenario-studies rather than many real-world datapoints at full gigawatt scale; while they reference existing clusters and faults, the true gigawatt-scale dynamic remains partly speculative. (They model e.g., West Texas datacenter load disconnection risk)

The cost and engineering feasibility of deploying these buffering/hardware solutions at the required scale remain uncertain — e.g., deploying synchronous condensers or large battery banks for multiple GW loads is expensive and complex.

The article focuses primarily on training workloads (which are highly synchronous) and less on inference workloads (though they mention inference could become more challenging) — so there is a potential blind-spot as inference scale grows.

Finally, there is a limitation in forecasting: predicting exactly how many GW of AI load will connect where and when, and what their dynamic profiles will look like, remains highly uncertain.

Future Research Directions

More detailed empirical measurement of AI datacenter power-profile dynamics at scale: e.g., quantifying the milliseconds/seconds fluctuations for large clusters, and capturing real-world data rather than just modelling.

Improving interconnection and grid-planning studies to explicitly incorporate AI-dedicated large loads with rapid dynamic behaviour (ramp/rate of change) in grid stability models.

Investigating advanced buffering / storage technologies (e.g., supercapacitors, fast-response BESS) tailored to the unique load fluctuations of AI systems.